

OMAR FADLALLAH

AI Engineer | LLM Systems | RAG Architectures | NLP

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PROFESSIONAL SUMMARY

AI Engineer who started in Data Analysis and BI before moving deliberately into ML and production GenAI — mastering the data foundation first, then scaling into agentic systems. Built enterprise-grade RAG pipelines achieving **sub-50ms P50 latency**, a multi-agent AI Operating System deployed on Hugging Face Spaces, and a **70% CER reduction** on Arabic legal OCR through fine-tuning a 4B vision-language model. Interned at Devoteam delivering enterprise AI PoCs; **NVIDIA DLI-certified** in Applied Deep Learning and Gen AI.

TECHNICAL SKILLS

LLM & RAG: Retrieval-Augmented Generation, Prompt Engineering, LLM Fine-tuning (LoRA / QLoRA), Agentic Systems, Hybrid Retrieval, Cross-encoder Re-ranking, Multi-step Reasoning

Frameworks: LangChain, LangGraph, HuggingFace Transformers, PyTorch, FastAPI, Gradio

Vector & Search: Qdrant, FAISS, BM25, Semantic Search, Redis (semantic cache)

Evaluation & MLOps: MLflow, Prometheus, Grafana, Celery, ZenML, Docker

Serving: vLLM, ONNX Runtime

Cloud & DevOps: AWS, Hugging Face Spaces, GitHub Actions

Data & Storage: PostgreSQL, MySQL, MongoDB, Redis, Pandas, NumPy, SQLAlchemy

Languages: Python (primary), SQL, C++, R • Arabic (native), English (fluent)

Certifications: NVIDIA DLI – Applied Deep Learning • NVIDIA DLI – Generative AI • AWS Practical Training (DataCamp)

PROFESSIONAL EXPERIENCE

Analytics & Agentic AI Intern • [Kayfa](#)

May 2025 – Jun 2025

- Engineered and deployed a Hybrid RAG Course Advisor Agent (FastAPI, PydanticAI, LlamaIndex, MongoDB, Groq) with semantic retrieval, streaming responses, and JWT-secured conversation management — live on Hugging Face Spaces. [Live Demo](#)
- Built a Student Analytics Dashboard (Streamlit, MongoDB Atlas, Plotly) aggregating 500+ students from 8 data sources, with a 41-point ETL audit and a K-Means risk-scoring model for 4-tier student segmentation. [Live Demo](#)
- Automated CI/CD pipelines (GitHub Actions) for multi-container, zero-downtime deployments across Hugging Face Spaces.

AI Engineer Intern • [Devoteam](#)

Dec 2025 – Feb 2026

- Fine-tuned BERT-based models (PyTorch) for text classification and summarization, improving F1 by ~15% over baselines on 10,000+ labeled samples.
- Built RAG ingestion pipelines processing 50,000+ enterprise document chunks into a vector store, improving retrieval latency and precision for downstream LLM responses.
- Optimized prompt engineering for 2 enterprise AI PoCs, improving answer groundedness against client-defined evaluation rubrics.

KEY PROJECTS

Computational Drug Discovery & Repurposing Platform ★ *Graduation Project* — **Lead AI & Backend Engineer** • **Team of 9**

Flask • FAISS-IVF • RDKit • Chemprop • PyTorch Lightning • DeepPurpose • OpenMM • AmberTools • PyTraj • Docker • asyncio • Celery • Redis

- Architected a production-grade drug discovery platform unifying Chemical RAG, ADMET prediction, drug repurposing, and molecular dynamics simulation into a single modular, async backend.
- Chemical RAG at Scale: Indexed 1M+ compounds in a FAISS-IVF vector store with sub-100ms similarity retrieval; grounded Llama-3.1-8B with RDKit-extracted molecular properties to enable reliable scientific reasoning without fine-tuning.
- ADMET Optimization (10s → 150ms): Trained 5 task-specific MPNN models from scratch on molecular property data, then engineered a concurrent, CPU-optimized inference engine — combining async request handling with parallel execution — to serve them. Cut single-molecule prediction latency from 10s to 50–150ms with no GPU dependency.
- Drug Repurposing Pipeline: High-throughput screening of 5,000+ approved drugs against protein targets using binding-affinity models, with automated external API enrichment.
- Molecular Dynamics: Productized a multi-step simulation protocol into a single autonomous API, offloading long-running jobs to Celery/Redis workers with real-time status polling; deployed via multi-stage Docker with GitHub Actions CI/CD.

Atlas AI Platform — **Enterprise Multi-Tenant RAG** — [GitHub](#) | [Demo](#) | *Python • LangChain • LangGraph • Qdrant • FAISS • Redis • FastAPI • Docker • Prometheus • Grafana*

- Designed a multi-tenant SaaS RAG platform with JWT authentication and strict per-tenant data isolation, enabling secure document Q&A for multiple organizations on a shared infrastructure.
- Built hybrid retrieval pipeline combining dense vector search (Qdrant), BM25 keyword matching, and cross-encoder re-ranking — improving answer relevance and reducing hallucinations compared to single-retriever baselines.

- Implemented a 3-tier caching architecture (in-memory → Redis semantic cache → database) achieving sub-50ms P50 API latency under load and cutting LLM API call volume by 60–80% through semantic cache hit reuse.
- Fine-tuned Qwen2.5-1.5B with LoRA/QLoRA on Arabic instruction-following data using 4-bit NF4 quantization, improving domain-specific Arabic Q&A accuracy while keeping inference cost minimal on consumer hardware.
- Added per-request cost and token tracking with real-time observability via Prometheus and Grafana (latency, token usage, drift alerts).
- Built a LangGraph-based agentic reasoning layer supporting multi-step query decomposition and tool use for complex questions.

Arabic Legal OCR — Fine-tuning Vision-Language Models for Structured Extraction — [GitHub](#) | [PyTorch](#) · [LlamaFactory](#) ·

[LoRA/QLoRA](#) · [vLLM](#) · [Gemma-3-4b-it](#) · [HuggingFace PEFT](#)

- Fine-tuned Gemma-3-4b-it (4B params) using LoRA on Arabic legal documents, reducing Character Error Rate from 0.74 to 0.22 — a 70% CER improvement — and lifting OCR accuracy from ~26% to ~78%.
- Scraped and processed 72+ legal documents from the Saudi Ministry of Justice API; used knowledge distillation with Gemini 1.5 Pro to generate structured JSON training labels.
- Built a dual-task extraction pipeline outputting structured document content and metadata (issuing authority, dates, signatures) with confidence scoring.
- Deployed production inference via vLLM with multi-LoRA support; trained with 4-bit NF4 quantization, enabling highly efficient fine-tuning on a consumer-grade GPU with zero performance loss.

AI-lirix — Multi-Agent Scientific Operating System — [Live Demo](#) | [FastAPI](#) · [LlamaIndex](#) · [Groq](#) · [Weaviate](#) · [Docker](#) · [HuggingFace Spaces](#)

· [WebSocket](#) · [Whisper](#)

- Architected a multi-agent AI backend for drug discovery routing natural language queries (text and voice) to specialist agents — Chemical (ADMET, SMILES similarity, drug repurposing), Medical (biomedical LLM reasoning), RAG (documentation Q&A), and Conversational — via a combined Groq LLM orchestrator with keyword fallback classifier.
- Built a hybrid RAG pipeline (vector + BM25) using LlamaIndex with multilingual embeddings for accurate Arabic and English retrieval; implemented three-tier persistence with additive ingestion to prevent data loss on reload.
- Delivered a real-time voice-to-voice agent pipeline: WebSocket channel with VAD, Groq Whisper STT, streaming AI response tokens, and OpenAI TTS — supporting interruption mid-stream and bilingual (Arabic/English) scientific transcription.
- Engineered production deployment on HuggingFace Spaces (Docker): resolved critical dependency conflicts, embedding reload bugs, and Redis cold-start issues; added an in-process monitoring dashboard and GitHub Actions CI/CD sync.

LLMOps Platform — Production Prompt Management System — [GitHub](#) | [Demo](#) | [Python](#) · [FastAPI](#) · [React/Vite](#) · [Celery](#) · [PostgreSQL](#)

· [Docker](#) · [Redis](#) · [LangChain](#)

- Built a full-stack LLMOps platform (FastAPI + React/Vite) to prevent silent prompt regressions and untracked LLM costs in production — with prompt versioning, API key authentication, and Redis-backed rate limiting (60 req/min per key).
- Designed a regression-testing engine that runs every prompt version against a fixed set of golden examples, using an LLM-as-judge evaluator to score outputs and flag hallucinations before they reach production.
- Implemented per-run cost, token, and latency tracking alongside prompt version diffing and one-click rollback, giving full visibility into what changed and what it cost.
- Offloaded long-running LLM calls and batch evaluations to Celery/Redis workers for async, non-blocking execution; data layer uses PostgreSQL/SQLAlchemy with Alembic migrations.

EDUCATION

B.Sc. Computer Science · [Faculty of Computers and Information, Mansoura University](#)

Oct 2022 – Jun 2026

GPA: 3.7 / 4.0 · Top 5% of cohort · Graduated: June 2026 · Military Status: Exempted